HOIP Project

Last Update: @March 3, 2026 4H-Mn-BA Sample 4 is measuring under SCXRD.

The measurement database is structured with each entry representing a distinct sample, including all associated properties and characterizations. Each sample is linked to a chemical short name, enabling efficient filtering of samples associated with the same chemical.

Since @May 24, 2025, A sample Inventory is added. As a exhausted list of all of samples, it serves to reduce the risk of suddenly run out of sample within experiment.

Since @November 12, 2025, we have received the newest batch of sample from our Korea. From now on, we are only using the newest batch of sample.

In the supplementary resources, there's some links to the Google Slides and Google sheets, where you could see more graphs.

Table of Contents

- Overview of Data Collection
- Crystal Inventory
- Lattice Parameters of Group Mn-BA
- Representative Graph for each Single Crystal Sample（PPMS）
  - Mn-PEA
    - VSM
    - Heat Capacity
    - Phase Diagram
  - 4Cl-Mn-BA
  - 4H-Mn-BA
  - 4F-Mn-BA
  - 4Br-Mn-BA
  - 4I-Mn-BA
- Experimental Log
- Remaining Tasks
- Supplementary Resources
  - Slides and Sheets
- Related Pages

Measurement Database

Sample Name	Make Date	Sample Type	Hysteresis	HeatCapacity	DC-Magnetisim	# Mass (mg)
4Br-						
4Cl-Mn-BA - 5	19/11/2025	Single Crystal - OOP	2P5K		0P01T_FC 0P01T_ZFC 0P1T_FC 0P1T_ZFC	12.84
4I-Mn-BA - 2	19/11/2025	Single Crystal - OOP			0P01T_ZFC	7.54
4Br-Mn-BA - 3	19/11/2025	Single Crystal - OOP			0P01T_ZFC	6.57
4H-Mn-BA - 5	18/11/2025	Single Crystal - OOP	2P5K		0P01T_FC 0P01T_ZFC 0P1T_FC 0P1T_ZFC	9.82
4Cl-Mn-BA - 4	18/11/2025	Single Crystal - OOP			0P01T_ZFC	12.43
4I-Mn-BA - 1	18/11/2025	Single Crystal - OOP			0P01T_ZFC	8.89
4Br-Mn-BA - 2	17/11/2025	Single Crystal - OOP			0P01T_FC 0P01T_ZFC 0P1T_FC 0P1T_ZFC	9.69
4Cl-Mn-BA - 3	17/11/2025	Single Crystal - OOP			0P01T_ZFC	10.85

Sample Name	Make Date	Sample Type	Hysteresis	HeatCapacity	DC-Magnetisim	# Mass (mg)
4H-Mn-BA - 4	17/11/2025	Single Crystal - OOP			0P01T_FC	8.09
4H-Mn-BA - 3	13/05/2025	Single Crystal		0Oe 1000Oe	0P1T_ZFC	1.55
4Cl-Mn-BA - 4	02/05/2025	Single Crystal	2P5K	0Oe 100Oe	0P01T_FC 0P01T_ZFC 0P1T_FC 0P1T_ZFC	1.88
Mn-PEA - 3	03/04/2025	Single Crystal		1000Oe	0P1T_ZFC	7.73
4Cl-Mn-BA - 3	18/03/2025	Single Crystal		0Oe 1000Oe 100Oe	0P1T_ZFC	2.87
4Br-Mn-BA - HC	06/03/2025	Single Crystal		0Oe		0.33
Untitled						
4H-Mn-BA - 2	06/02/2025	Single Crystal - OOP	2P5K		0P01T_FC 0P01T_ZFC 0P1T_FC 0P1T_ZFC 14T_FC	3.46
4H-Mn-BA - 2	05/02/2025	Single Crystal - IP	2P5K		0P01T_FC 0P01T_ZFC 0P1T_FC 0P1T_ZFC 14T_FC	10.36
4F-Mn-BA -2	04/02/2025	Single Crystal - IP	2P5K		0P01T_FC 0P01T_ZFC 0P1T_FC 0P1T_ZFC	0.5
4Br-Mn-BA - 1	31/01/2025	Single Crystal - OOP	2P5K		0P01T_FC 0P01T_ZFC 0P1T_FC 0P1T_ZFC 1P4T_FC	4.22
4Br-Mn-BA - 1	30/01/2025	Single Crystal - IP	2P5K		0P01T_FC 0P01T_ZFC 0P1T_FC 0P1T_ZFC 1P4T_FC	9.28
4F-Mn-BA - 1	29/01/2025	Single Crystal - OOP	2P5K		0P01T_FC 0P01T_ZFC	1.2

Sample Name	Make Date	Sample Type	Hysteresis	HeatCapacity	DC-Magnetisim	Mass (mg)
					0P1T_FC 0P1T_ZFC 1P4T_FC	
4F-Mn-BA - 1	28/01/2025	Single Crystal - IP	2P5K		0P01T_FC 0P01T_ZFC 0P1T_FC 0P1T_ZFC	3.06
4F-Mn-BA - 1	10/12/2024	Single Crystal - IP				4.31
4H-Mn-BA - 1	09/12/2024	Single Crystal - OOP	2P5K		0P01T_FC 0P01T_ZFC 0P1T_FC 0P1T_ZFC 14T_FC	4.33
4H-Mn-BA - 1	06/12/2024	Single Crystal - IP	2P5K		0P01T_FC 0P01T_ZFC 0P1T_FC 0P1T_ZFC 14T_FC	9.05
Mn-PEA - 2	02/12/2024	Fine Powder				11.18
4Cl-Mn-BA - 2	18/10/2024	Single Crystal - OOP	2P5K 35K 45K		0P01T_FC 0P01T_ZFC 0P1T_FC 0P1T_ZFC 14T_FC	42.96
4Cl-Mn-BA - 2	16/10/2024	Single Crystal - IP	2P5K		0P01T_FC 0P01T_ZFC 0P1T_FC 0P1T_ZFC 14T_FC	75.59
4F-Mn-PEA - 3	03/10/2024	Fine Powder	2P5K		0P01T_FC 0P01T_ZFC 0P1T_FC 0P1T_ZFC 0P5T_ZFC	10.74
4Cl-Mn-BA - 1	02/10/2024	Single Crystal - OOP	2P5K		0P01T_FC 0P01T_ZFC 0P1T_FC 0P1T_ZFC 14T_FC	18.88

Sample Name	Make Date	Sample Type	Hysteresis	HeatCapacity	DC-Magnetisim	Mass (mg)
4Cl-Mn-BA - 1	01/10/2024	Single Crystal - IP	2P5K		0P01T_FC 0P01T_ZFC 0P1T_FC 0P1T_ZFC 14T_FC	20.13
4F-Mn-PEA - 2	30/09/2024	Fine Powder	2P5K		0P01T_FC 0P01T_ZFC 0P1T_FC 0P1T_ZFC	8.43
4F-Mn-PEA - 1	27/09/2024	Coarse Powder	2P5K 39K 60K		0P01T_FC 0P01T_ZFC 0P05T_FC 0P05T_ZFC 0P1T_FC 0P1T_ZFC 0P5T_ZFC 10T_FC 10T_ZFC 1P4T_FC 1P4T_ZFC	9.78
Mn-PEA - 1	15/07/2024	Single Crystal - OOP	10K 15K 20K 25K 2P5K 30K 32K 34K 36K 38K 40K 41K 42K 43K 44K 45K 5K 60K		0P001T_ZFC 0P01T_ZFC 0P1T_ZFC 10T_ZFC 1T_ZFC	9.12

Overview of Data Collection

Progress Overview	Sample Screening	X-ray Diffraction	Phase Diagram Mapping	
			In Plane	O
4Cl-Mn-BA	✅	⚠️	⚠️	⚠
4Br-Mn-BA	✅	✅	✅	🔴
4H-Mn-BA	⚠️	⚠️	⚠️	⚠
4I-Mn-BA	🔴	🔴	⚠️	⚠
4F-Mn-BA	🔴	🔴	🔴	🔴

✅ : Data complete and usable

⚠️ : Data either partially exist, or needs improve.

🔴 : Data not yet collected or unusable

Crystal Inventory

Mn-BA Sample Batch Nov 2025

Aa Name	Chemical	Box Numebr
Untitled		

Lattice Parameters of Group Mn-BA

Link to CIF Files: https://seafile.xiaomie-cloud.de/d/d21b0f23e4d54b4abbe9/?p=%2F&mode=list

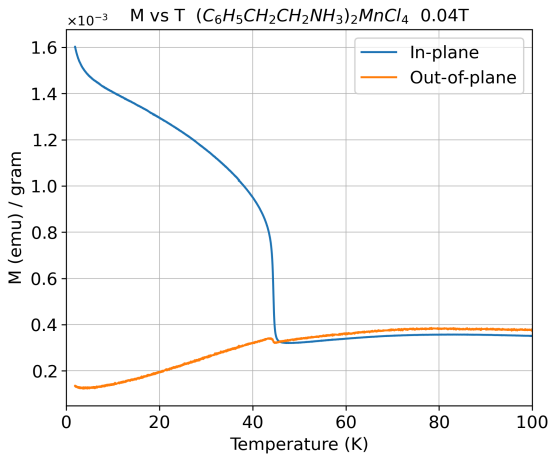
Chemical Formula	4Cl-Mn-BA	4Br-Mn-BA	4H-Mn-BA	4F-Mn-BA	4I-Mn-BA
Space Group	C1c1 Monoclinic	C1c1 Monoclinic	C1c1 Monoclinic	Pbnm orthorhombic	Pbnm orthorhombic
	Polar	Polar	Polar	Non polar	Non polar
a	34.40	35.15	32.5301	5.15	5.15300
b	5.18	5.19	5.1656	10.41	10.41400
c	10.41	10.46	10.3365	32.05	32.05100
α	90	90	90	90	90
β	98.61	98.47	98.37	90	90
γ	90	90	90	90	90

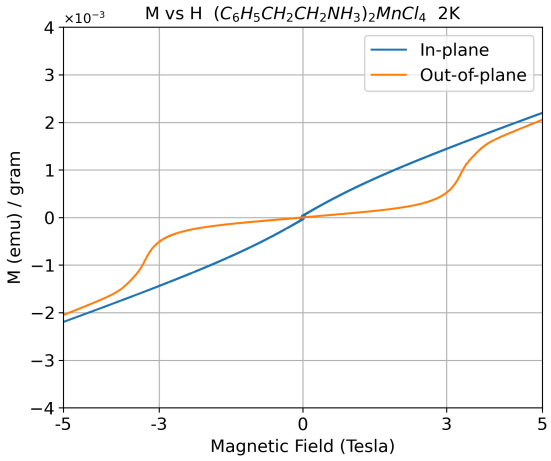
Representative Graph for each Single Crystal Sample (PPMS)

Mn-PEA

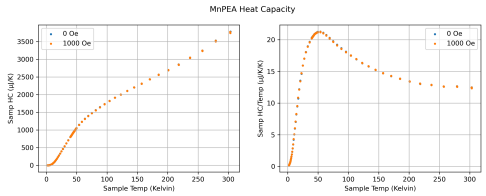
VSM

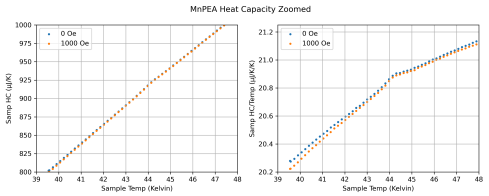
The Data is from Lukas's old sample. DSC4 and DSC6.



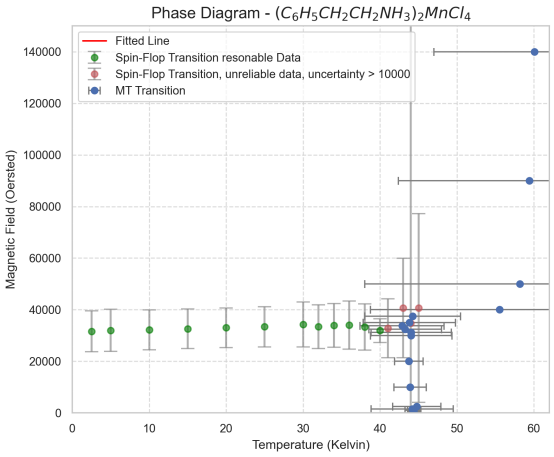


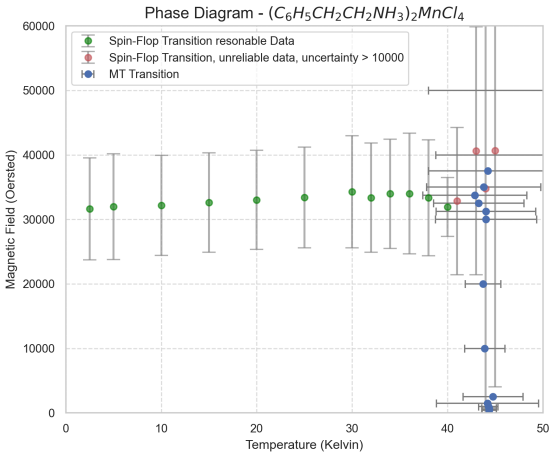
Heat Capacity





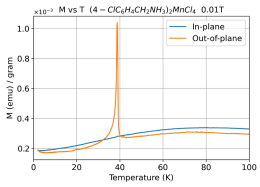
Phase Diagram

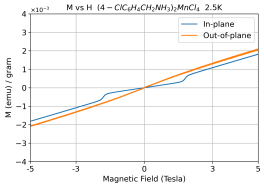


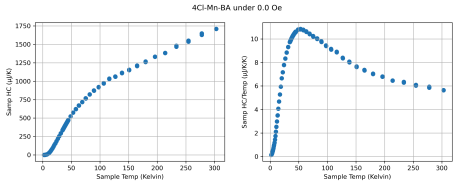


4Cl-Mn-BA

This is represented by our first sample of this chemical. 4Cl-Mn-BA - 1

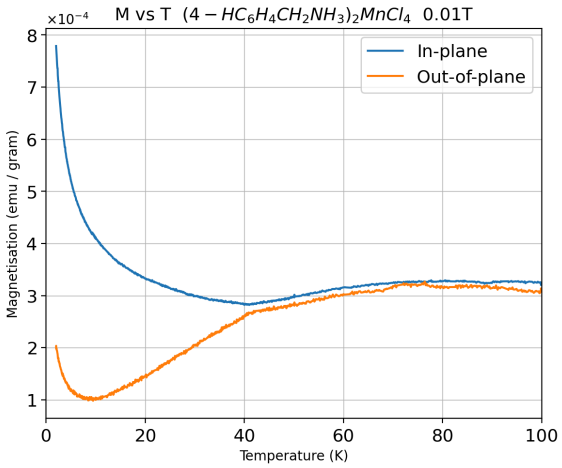


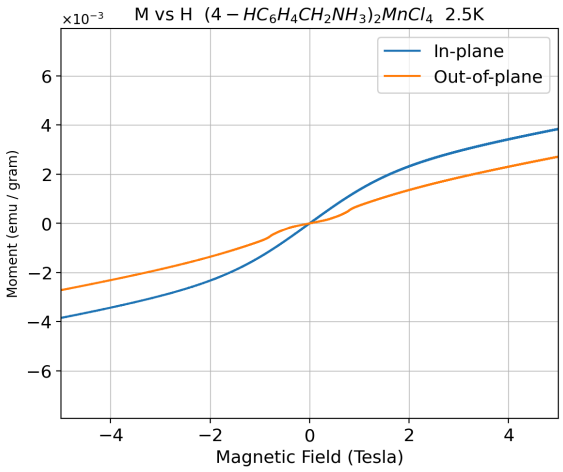


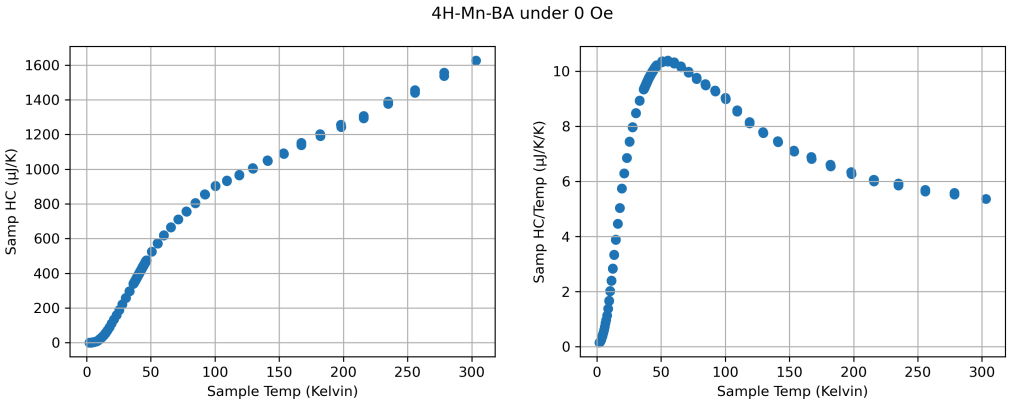


4H-Mn-BA

This is represent by sample 4H-Mn-BA - 1

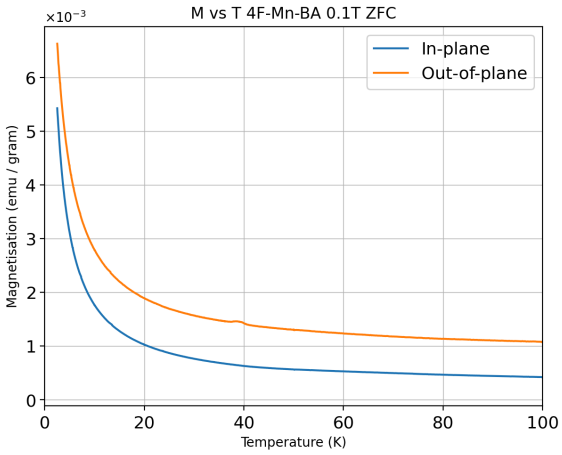


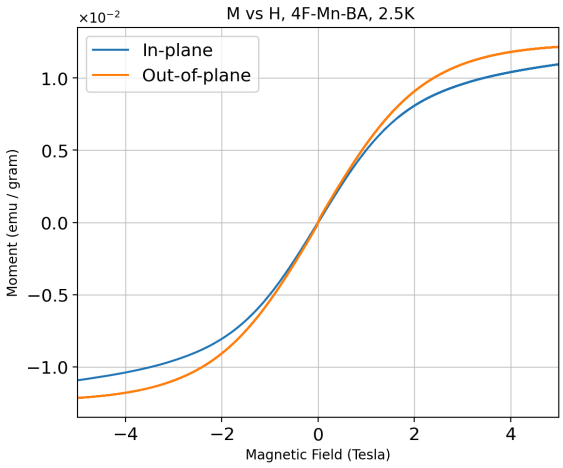




4F-Mn-BA

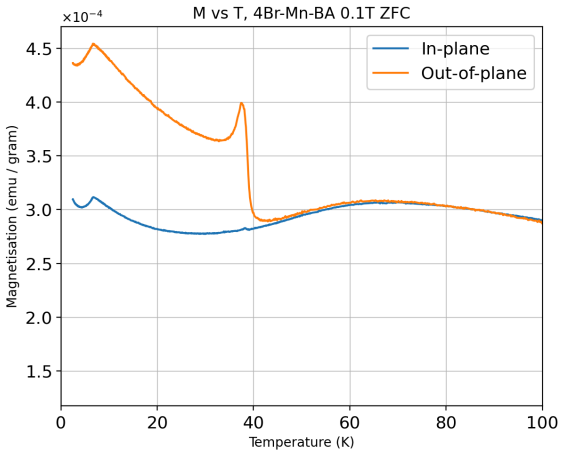
This is represent by sample 4F-Mn-BA - 1

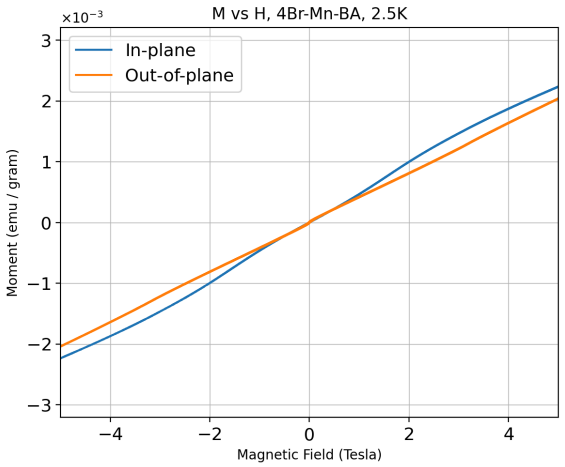


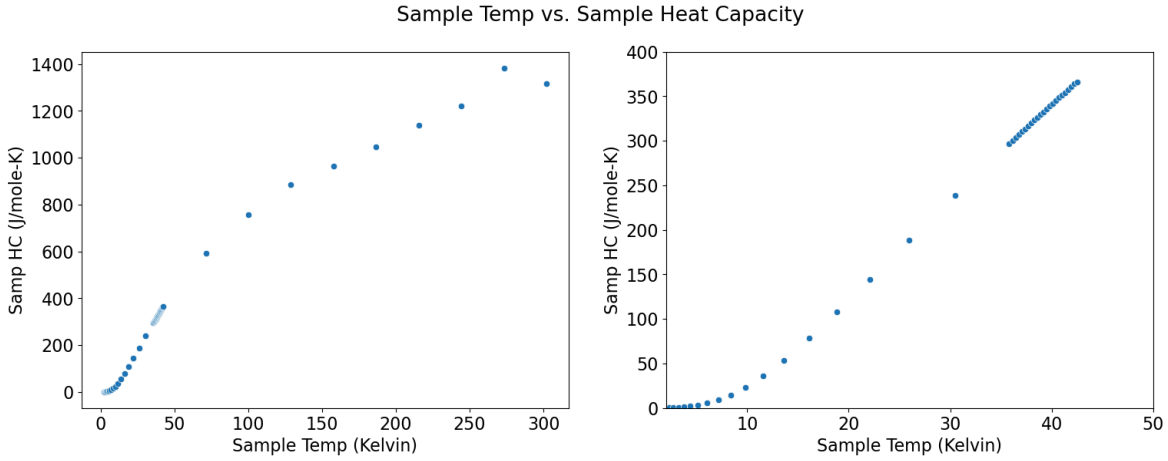


4Br-Mn-BA

This is represent by sample 4Br-Mn-BA - 1







4I-Mn-BA

This is represent by sample 4I-Mn-BA - 1

Experimental Log

You might find more information in the experiment log. 📕 Experiment Log

Remaining Tasks

Supplementary Resources

Slides and Sheets

1. Slides on the Heat Capacity of 4Br-Mn-BA(https://docs.google.com/presentation/d/1ucPJPpWxygOMiFvwQmkmXhFThkssXKgHPdDIbdq89Fg/edit?

usp=sharing)

2. Slides on the Preliminary Data Analysis of Mn-BA Family. (https://docs.google.com/presentation/d/14CqMsGa7EKI_4Rs4NmvhKd-Wn6MhK4c5MM-ijIkcgoc/edit?usp=sharing)
3. Slides on Phase Transition. https://docs.google.com/presentation/d/1X-4iGbIQJseDX860D4ec9EeJL5bwRK_0QAYeuVpTyuY/edit?usp=sharing
4. Slides on Spin Models. https://drive.google.com/file/d/12iQF3_1eWhgw33XQn2c9i1TAKk23ixcw/view?usp=sharing

Related Pages

- Sample Inventory Old
- Experiment Log